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Points to Consider No. 13

Materials in ATMP Manufacturing

2025

Parenteral Drug Association®

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Introduction

This document provides key "Points to Consider" for navigating the complexities of raw materials in the manufacturing of advanced therapy medicinal products (ATMP), aiming to guide stakeholders in establishing robust strategies for raw material management, promoting best practices, and supporting the safe and effective development and manufacturing of ATMPs. The unique nature of ATMPs presents distinct challenges compared to traditional pharmaceuticals, demanding a focused approach to material selection, qualification, and supplier management.

Specifically addressed in this document are the intricacies arising from the fact that ATMP raw materials are often animal- or human-derived, not monographed, not manufactured for good manufacturing practice (GMP) applications (e.g., research-only grade), single-sourced and/or required to be used for aseptic manufacturing.

Very often, ATMP material suppliers and vendors are small, specialized companies with roots in academia that do not have a mature GMP quality system.

These characteristics contribute to a complex regulatory landscape, as regulations for ATMP materials are not globally harmonized. Variations in regulations exist among the U.S. Food and Drug Association (FDA), European Medicines Agency (EMA), Japan Pharmaceutical and Medical Devices Agency (PMDA), and the Pharmaceutical Inspection Co-operation Scheme (PIC/S), as well as individual country-specific laws to which companies must adhere.

Scope

This document applies to all process-stream-contacting raw materials, including starting and ancillary materials, excipients, consumables, and container–closure systems for ATMPs both in clinical stage and commercial manufacture. This includes but is not limited to cell-free mRNA therapeutics, viral vector-based in-vivo gene therapies, human cells for autologous or allogeneic cell therapies, or tissue-engineered medicinal products.

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Glossary

Definitions have been provided to help clarify the concepts discussed in this document. While some definitions used vary among companies, geographic location, etc., the definitions described below are consistently used within this Points to Consider document. Where a definition is based on another published source, the source is cited.

Creutzfeldt-Jakob Disease
A rare, fatal brain disorder belonging to a group of diseases called prion diseases. It is caused by an abnormal infectious protein, a prion, that accumulates in the brain, leading to rapid mental decline and neurological symptoms.

Primary Animal-Derived Materials
Any material derived directly from animal sources.

Primary Human-Derived Materials
Any material derived directly from human sources.

Secondary Animal-Derived Materials
Any material manufactured using processes that include materials derived from animal sources.

Secondary Human-Derived Materials
Any material manufactured using processes that include materials derived from human sources.

Tertiary Animal-Derived Materials
Any material free of animal-derived components that has been manufactured using secondary animal-derived materials.

Tertiary Human-Derived Materials
Any material free of human-derived components that has been manufactured using secondary human-derived materials.

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Abbreviations

AAV	Adeno-associated Virus
ATMP	Advanced Therapy Medicinal Products
BLA	Biologics License Application
CAR	Chimeric Antigen Receptor
CFR	Code of Federal Regulations
CJD	Creutzfeldt-Jakob Disease
CoA	Certificate of Analysis
CoO	Certificate of Origin
gRNA	guide RNA (ribonucleic acid)
hESC	human Embryonic Stem Cell
IND	Investigational New Drug
iPSC	induced Pluripotent Stem Cells
JSBRM	Japanese Standards for Biological Raw Materials
PBMC	Peripheral Blood Mononuclear Cells
RUO	Research Use Only
TIL	Tumor Infiltrating Lymphocyte
TSE	Transmissible spongiform encephalopathies
BSE	Bovine spongiform encephalopathy

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Topic 1. ATMP Material Categories

Q 1.1. Which ATMP material categories exist and how are they defined?

There is no health authority guideline that clearly defines ATMP material categories nor an aligned definition across countries or regions. The material categories listed in **Table 1.1-1** reflect existing regulatory guidance relevant to ATMPs and common industry vernacular.

The term "critical raw material" is used by European Union (EU) health authorities to describe materials critical for product quality in their *Guidelines on Good Manufacturing Practice Specific to Advanced Therapy Medicinal Products* ***(1)***. This guideline does not provide a clear definition of a "critical raw material," however, but lists a few examples such as sera, growth factors, enzymes (such as trypsin), cytokines, and gene-editing reagents). ATMP sponsors may define the criticality of raw materials based on their specific products and internal procedures.

The term "material" is a general term used to describe manufacturing ingredients, raw materials (e.g., starting and ancillary materials, excipients, primary packaging) consumed in the process that may or may not be present in the final drug product ***(2)***. The material categories given in **Table 1.1-1**, as well as "critical raw materials," fall under "materials."

The term "animal-derived raw material" or "animal-derived material" is widely used, but its definition has not yet been globally aligned by health authorities. PDA *Technical Report No. 83: Virus Contamination in Biomanufacturing: Risk Mitigation, Preparedness, and Response* and *European Pharmacopoeia*, *Chapter 5.2.12: Raw Materials for the Production of Cell-Based and Gene Therapy Medicinal Products*, provide definitions ***(3, 4)***, which the authors consolidated into the following:

Primary Animal-Derived Materials

Any material derived directly from animal sources

Examples

- Bovine serum
- Porcine-derived trypsin
- Animal-tissue-derived hydrolysates

Secondary Animal-Derived Materials

Any material manufactured using processes that include materials derived from animal sources

Examples

- Recombinant protein produced in an *E. coli* fermentation that uses a medium containing bovine tryptone
- Recombinant protein expressed in plants that are exposed to bovine manure fertilizer

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- Recombinant CHO-cell culture products

Tertiary Animal-Derived Materials

Any material free of animal-derived components that has been manufactured using secondary animal-derived materials

Examples

- Tubing sets or other polymeric components that were manufactured using slip agents derived from animal fat (stearates)
- Solvent that has been purified using bone-derived, activated charcoal

The definitions provided above for primary and secondary animal-derived materials are widely applicable to human-derived (raw) materials as well.

Primary Human-Derived Materials

Any material derived directly from human sources

Examples

- Human donor cells
- Human serum
- Human Serum Albumin

Secondary Human-Derived Materials

Any material manufactured using processes that include materials derived from human sources

Examples

- Recombinant protein produced in a human cell line

Tertiary Human-Derived Materials

Any material free of human-derived components that has been manufactured using secondary human-derived materials

Any of the material categories presented in **Table 1.1-1** could also be subclassified as primary, secondary, or tertiary animal-derived or primary, secondary, or tertiary human-derived mater

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Table 1.1-1 Commonly Used ATMP Material Categories

Material Category	Definition	Reference Document	Examples
Ancillary Raw Materials	Raw materials used during the medicinal product processing that are not intended to be a part of the final product formulation	Modified from: USP <1043> ISO 20399	• Feeder cells • Growth factors • Cell differentiation reagents • Expansion Media

Starting Materials	ATMP starting materials are defined as all the materials from which the active substance is manufactured	EMA/CAT/22473/2025
		
		EMA/246400/2021
			Based on health authority feedback, the following ATMP starting materials have been defined:
			• mRNA-based individualized therapy:
			  ○ DNA template for in-vitro mRNA transcription
			  ○ mRNA capping reagent
			  ○ RNA ribonucleotides
			• AAV-based gene therapy:
			  ○ Plasmids
			  ○ Cell banks (e.g., HEK293)
			• CAR T-cell therapy:
			  ○ Patient leukapheresis material
			  ○ Cell Banks
			  ○ Packaging and transfer plasmids and plasmid cell banks
			  ○ Viral and non-viral vectors
			  ○ Gene editing reagents (e.g., gRNA, CAS nuclease)
			• TIL therapy:
			  ○ Tumor specimen

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Excipients	An excipient is any ingredient or substance intentionally added to a drug that is not part of the active substance (e.g., to aid in the protection, support, stability, bioavailability, or patient acceptability).	EMEA/CHMP/QWP/396951/2006, FDA 2005 Guidance for Industry Nonclinical Studies for the Safety Evaluation of Pharmaceutical Excipients	• HEPES–liposomes and lipid nanoparticles for mRNA therapeutics • DMSO–biological preservative like human serum, and proprietary materials for cell therapies
Subcategory of Excipients– Novel Excipient	A novel excipient is one used for the first time in a drug product or by a new route of administration. It may be a new chemical entity or a well-established one that has not yet been used for human administration and /or for a particular human administration pathway.	EMEA/CHMP/QWP/396951/2006, FDA 2005 Guidance for Industry Nonclinical Studies for the Safety Evaluation of Pharmaceutical Excipients	PF68 in AAV formulation for intra-cisternal magna injection
Single-Use Components	Components directly contacting the process stream in product manufacturing for a single purpose and discarded	BioPhorum 2019 Raw Material Risk Assessments A holistic approach to Raw Material Risk Assessments through Industry Collaboration	• Bioprocess bags • Tubing • Filters • Connectors • Microcarriers • Magnetic beads
GMP Support Materials	Procured materials supporting GMP manufacturing without direct product contact	BioPhorum 2019 Raw Material Risk Assessments A holistic approach to Raw Material Risk Assessments through Industry Collaboration	• Clean-in-place • Cleaning agents/disinfectants • Aseptic gowning materials • Process equipment gaskets • Process-simulation media

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Laboratory Reagents (Testing Materials)	Used as part of analytical testing either in-line, in-process or off-line, with no contact with the process stream	BioPhorum 2019 Raw Material Risk Assessments A holistic approach to Raw Material Risk Assessments through Industry Collaboration	• Analytical reference standards, assay controls and materials • Assay-specific critical reagents • Cell lines • Positive control test material
Primary Packaging Components/Container–Closure Systems	Container–closure systems and device components directly in contact with the final drug product or directly employed for the delivery of the final drug product	USP <659>	• Cryo bags • Vials • Stoppers • Caps • Prefilled syringes
Process Aids	Materials used to facilitate the manufacturing process that are not consumed during processing and may or may not be multiuse (primarily a physical function)	BioPhorum 2019 Raw Material Risk Assessments A holistic approach to Raw Material Risk Assessments through Industry Collaboration	• Resins • Chromatography columns • Process filters • In-process containers
Process Gasses	Gases that come in direct contact with the product or primary container surfaces	The Rules Governing Medicinal Products in the European Union Volume 4 EU Guidelines for Good Manufacturing Practice for Medicinal Products for Human and Veterinary Use Annex 1 Manufacture of Sterile Medicinal Products	• Material transfer ports integrated into rigid barrier

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Topic 2. Defining an ATMP Material Strategy

Q 2.1. What are key considerations when defining an ATMP material strategy?

Defining an ATMP material strategy means to constantly maintain a balance between imperfections regarding material quality and supply standards and potential fast-track commercialization (e.g., clinical Phase 2 as a pivotal trial). This requires a risk-based approach that is both comprehensive and phase-appropriate and considers all aspects of material quality, including but not limited to manufacturer controls, contamination risk, and potential for process-related impurities. The objective is to always strive to maximize knowledge about the materials used at all stages. Focusing on the following key considerations has shown to be helpful.

Human- and Animal-Derived Materials

For many ATMPs (e.g., cell-based therapies), the use of animal-/human-derived materials is unavoidable during early clinical trials, and health authorities accept these materials if safety requirements for adventitious agents (e.g., microorganisms, viruses, prions) are met. However, sponsors are advised to work on replacing these materials in parallel with the product development, as they often pose a risk to eventual commercial manufacturing in terms of supply chain, batch-to-batch variability, and transmissible agents. It is important to note though that for some applications, no chemically defined replacements for animal-/human-derived materials have yet been found (e.g., human serum used for stem cell differentiation) as the development of these replacements requires extensive development activities.

Research Use Only Materials

Non-GMP material grades, such as research use only (RUO), can be acceptable for early clinical trials when higher-grade material is not openly available, but RUO material should be replaced with higher grades for later stages if possible. A potential replacement strategy could be to evaluate higher grades, once available, for their compatibility with the manufacturing process in parallel with the ongoing clinical manufacturing. Health authorities, such as EMA, may require companies to follow cGMP or GMP principles for starting materials and/or critical materials, and excipients for clinical phases and commercial manufacturing ***(5).***

Proprietary Materials

Certain key ATMP materials can be proprietary (e.g., cryoprotectants used as excipients, cell culture media), which complicates regulatory filing since sponsors do not have access to critical information associated with the intellectual property for the material (e.g., manufacturing method, full ingredients list). However, these materials are often not replaceable, and sponsors must find ways to mitigate the risk, for example, by submitting a Letter of Authorization to the FDA to access a supplier Master File.

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Chain of Identity/Chain of Custody

Manufacturers should have traceability systems in place to avoid material mix-ups and cross-contamination, as these could have severe consequences for patient safety. Especially for cell-based therapies, manufacturers must work with suppliers to ensure the chain of identity for human donor material in the entire supply chain (see **Topic 4**).

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Topic 3. Risk–Based Material Qualification

ATMP material qualification follows a risk-based approach (see *PDA Technical Report No. 81: Cell-Based Therapy Control Strategy* for risk assessment examples). Besides the general requirements, which apply to all clinical stages and commercial manufacturing, several requirements differentiate between early- and late-stage clinical to commercial transition. These differences must be considered when developing a comprehensive strategy that is also phase appropriate. Further, the sponsor of a clinical trial (manufacturer) is responsible for managing critical material information, like the Certificate of Analysis (CoA), Certificate of Origin, and material numbers, from material suppliers because health authorities require this information in regulatory filings to ensure traceability and consistent quality of material. Appropriate supplier management, along with phase-appropriate material testing, where required, are the key elements of qualification.

Q 3.1. What are key considerations for ATMP material qualification and specification setting?

Critical material attributes should be established based on the intended material use and final product quality. Material specifications are intended to ensure that materials meet the compliance requirements of the appropriate market and to ensure materials consistently meet the expected quality. Further, technical information regarding the composition, formulation, and characterization of critical materials is crucial for the assessment of suitability and safety. The level of risk related to the presence of organic and other impurities (e.g., nitrosamine, melamine, heavy metals, human allergens) must also be identified and evaluated. For ATMPs that fall under *PIC/S Guide to Good Manufacturing Practice for Medicinal Products regulation, Annex 2A, Part A, Section 5.23* provides additional guidance ***(6)***. For more information on the qualification and specification of plasmids and vectors, see *PDA Points to Consider No. 11: Development, Classification, Manufacture, Control, and Testing of Plasmids and Vectors Used in ATMP Production*.

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Q 3.2. What are the General Requirements?

Prior to setting up the manufacturing process and whenever a change of the respective material is implemented, a quality risk management (QRM) process should assess the overall risk to the entire manufacturing process and the resulting product.

Only materials that have been released by the Quality Unit that are within their expiration or retest date should be used.

Critical Material Attribute: Biological Safety

Ensuring patient safety and preventing transmission of infectious diseases is a general requirement for ATMP materials that applies to all clinical phases and commercial manufacturing.

Human- or animal-derived materials (see **Section 3.1** for a detailed definition) are tightly controlled by health authorities, as they pose the highest risk of transmitting pathogens to patients of all ATMP materials. The agreement between the ATMP manufacturer (or the Marketing Authorization Holder) and the supplier, including blood and tissue establishments, should contain clear provisions about the transfer of information. This should include the results of tests performed by the supplier, traceability data, and transmission of healthy donor information.

Controls can include, but are not limited to:

- Testing for human and animal viruses and other adventitious agents like mycoplasma and parasites.
- Virus inactivation (e.g., heat inactivation, low pH treatment, filtration, gamma-irradiation), if possible'
- Controlled sourcing and screening of human and animal materials to reduce the risk of CJD/TSE/BSE transmission.

Note: Based on their inherent risk profile, autologous cell therapies are usually exempted from donor (patient) testing, per 21 CFR 1271 ***(7).***

Many suppliers of ATMP materials cannot guarantee sterility assurance compliant with Eur. Ph. 5.1.1, EU Annex 1, EMA Guideline on the sterilization of the medicinal product, active substance, excipient and primary container, or similar regulations ***(8-10).*** Consequently, these raw materials should be considered as nonsterile even if they have been 0.2 µm-filtered. Sponsors may mitigate the risk associated with these nonsterile raw materials for aseptic ATMP manufacturing by considering the following aspects:

- Nonsterile materials should be added to the aseptic process via validated 0.2 µm filtration steps followed by a filter-integrity test. A generic filter validation per the ASTM F838-20 standard or equivalent is acceptable.
- Nonsterile raw materials should be tested for sterility (e.g., USP <71>) or compendial microbial enumeration testing (TAMC/TYMC) (e.g., USP <61>) with tight limits.

Table 3.2-1 provides an overview of relevant regulations and guidelines related to microbial and viral safety as well as TSE/BSE and CJD safety. This list is not exhaustive, and sponsors should also familiarize themselves with national laws that are required for the use of biological materials for ATMP manufacturing purposes. Health authority requirements can vary significantly and should be carefully assessed during early process development to avoid rejections or delays from the U.S. Investigational New Drug/European Investigational Medicinal Product Dossier (IND/IMPD) during the Biologics License Application (BLA)/Marketing Authorization Application process. Conducting a thorough gap

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assessment against relevant regulations is recommended to create a strategy for the product development lifecycle via consultation with health authorities for the intended region.

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Table 3.2-1 Overview of Relevant Guidelines Related to Microbial and Viral Safety and Transmissible Spongiform Encephalopathy/ Bovine Spongiform Encephalopathy and Creutzfeldt-Jakob Disease Safety

Type of ATMP Material	Relevant Guideline	Example	Comment
Non-Human/ Non -Animal Derived Material	U.S. GMP regulations EMA EudraLex – Vol. 4 for ATMPs	• Chemically defined cell culture media • *E. coli* plasmid • Nucleotides (e.g., guide RNA)	Microbial and viral safety requirements must be defined as part of the product's overall contamination control strategy
Human Blood Derived Material	• U.S. 21 CFR 610.40 • U.S. 21 CFR 640 • Ph. Eur. 5.2.12 • Ph. Eur. 1646 • MHLW Notification No. 37 (2018): Japanese Standards for Biological Raw Materials (JSBRM) • FDA Guidance for Industry: Recommendations to Reduce the Possible Risk of Transmission of Creutzfeldt-Jakob Disease and Variant Creutzfeldt-Jakob Disease by Blood and Blood Components (2022) • FDA Draft Guidance for Industry: Considerations for the Use of Human- and Animal-Derived Materials in the Manufacture of Cellular and Gene Therapy and Tissue-Engineered Medical Products (2024	• Human serum • Human serum albumin	None

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Human Tissue and Cell Derived Material	• FDA Guidance for Industry: Recommendations to Reduce the Possible Risk of Transmission of Creutzfeldt-Jakob Disease and Variant Creutzfeldt-Jakob Disease by Blood and Blood Components (2022) • U.S. 21 CFR 1271.50 • FDA Guidance for Industry: Chemistry, Manufacturing, and Control (CMC) Information for Human Gene Therapy Investigational New Drug Applications (INDs) (2020) FDA Guidance for Industry: Characterization and Qualification of Cell Substrates and Other Biological Materials Used in the Production of Viral Vaccines for Infectious Disease Indications (2010) • Ph. Eur. 5.2.12 • USP <1046> • EU Directive 2004/23/EC on setting standards of quality and safety for the donation, procurement, testing, processing, preservation, storage and distribution of human tissues and cells • Directive 2006/17/EC implementing -Directive 2004/23/EC • Directive 2002/98/EC on setting standards…for the collection, testing, processing, storage and distribution of human blood and blood components	• Feeder cells (e.g., cord-derived feeder cells) • HEK293 cell banks • Autologous: Apheresis material • Allogeneic: PBMC for iPSC Allogeneic: Cord blood • Allogeneic: hESC	Donor eligibility must be determined through screening for risk factors and/or clinical evidence of infection due to relevant communicable disease. Donor testing for relevant communicable disease is also required to demonstrate conformity with appropriate requirements.

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	• Directive 2004/33/EC implementing -Directive 2002/98/EC • Directive 2006/86/EC regards traceability requirements…for the coding, processing, preservation, storage and distribution of human tissues and cells • MHLW Notification No. 37 (2018): Japanese Standards for Biological Raw Materials (JSBRM) • "Enhancement of Quality and Safety of Drugs and Medical Devices Produced Using Raw Materials of Human Origin" (PFSB Notification No. 1213002		
Animal Derived	• 9 CFR 113: Standard Requirements —viral safety for animal materials • MHLW Notification No. 37 (2018): Japanese Standards for Biological Raw Materials (JSBRM) • ICH Q5A • WHO Guidelines on Transmissible Spongiform Encephalopathies in relation to Biological and Pharmaceutical Products 2003 • EMA/410/01	FBS	None

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Specific Safety Considerations for Allogeneic Therapies

For allogeneic therapies, testing at various levels (e.g., donor cells, cell banks, during drug substance/drug product manufacture) can address additional safety concerns from donor cells not covered by donor testing at the first timing of donation. Ensuring donor consent for commercial usage is recommended; otherwise, it may become an issue at the timing of commercialization if it is not fully agreed upon. For allogeneic ATMP manufacturing, which falls under the PIC/S GMP regulation, detailed guidance on seed lot and cell banking systems is given in *Annex 2a, Manufacture of Advanced Therapy Medicinal Products (ATMPs) for Human Use, Sections* 5.30 to 5.38.

ID Testing

Verification of identity is required for all raw materials that will be used to manufacture drug products; the requirements of global health authorities range from testing to the full specifications of the manufacturer to accepting on physical inspection and/or documentation. U.S. regulation 21 CFR 211.84(d)(2) requires a specific identity test, where available, to be performed for each lot of raw material received. In instances where specific tests are not available, surrogate testing to evaluate relevant characteristics (e.g. size, purity, spectral profile) may be combined to serve as verification of identity. EU regulation EudraLex - Vol. 4 allows acceptance of materials based on the manufacturer's CoA where all risks are understood or adequately mitigated by other means; however, it does recommend identity-testing for critical raw materials (e.g., excipients), see **Table 3.2-2** and **Figure 3.2-1** for an overview. The strategy for verifying the identity of raw materials upon receipt should be considered early, within the development phase of the product lifecycle.

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Table 3.2-2 Overview of United States and the European Union GMP Regulations Related to Material Identification Testing

Reference	Requirement	Guidance on Implementation
U.S. 21 CFR Part 210.2(c), Applicability of Current Good Manufacturing Practice Regulations	Production of an investigational drug for use in a Phase 1 study is exempt from compliance with the regulations in 21 CFR Part 211. However, this exemption does not apply once the investigational drug has been made available for use by or for the sponsor in a Phase 2 or Phase 3 study.	U.S. FDA Guidance for Industry: Investigational Drug for Use in a Phase 1 Study
U.S. 21 CFR Part 211.84(d)(2), Testing and Approval or Rejection of Components, Drug Product Containers, and Closures	Each component shall be tested for conformity with all appropriate written specifications for purity, strength, and quality. In lieu of such testing by the manufacturer, a report of analysis may be accepted from the supplier of a component, provided that at least one specific identity test is conducted.	USP <1043> Ancillary Materials for Cell, Gene, and Tissue-Engineered Products USP <1046> Cell-Based Advanced Therapies and Tissue-Based Products USP <1047> Gene Therapy Products USP <1665> Characterization and Qualification of Plastic Components and Systems Used to Manufacture Pharmaceutical Drug Products and Biopharmaceutical Drug Substances and Products
EudraLex - Vol. 4 - Guidelines on Good Manufacturing Practice Specific to Advanced Therapy Medicinal Products, Section 7.15	The ATMP manufacturer should verify compliance of the supplier's materials with the agreed specifications…Reliance on the certificate of analysis of the supplier is acceptable if all the risks are duly understood, and measures are put in place to eliminate the risks or mitigate them to an acceptable level (e.g. qualification of suppliers).	EMA Guideline on Quality, Non-Clinical and Clinical Aspects of Medicinal Products Containing Genetically Modified Cells EMA Guideline on Plasma-Derived Medical Products EMA Guideline on Quality, Non-Clinical and Clinical Requirements for Investigational Advanced Therapy Medicinal Products in Clinical Trials

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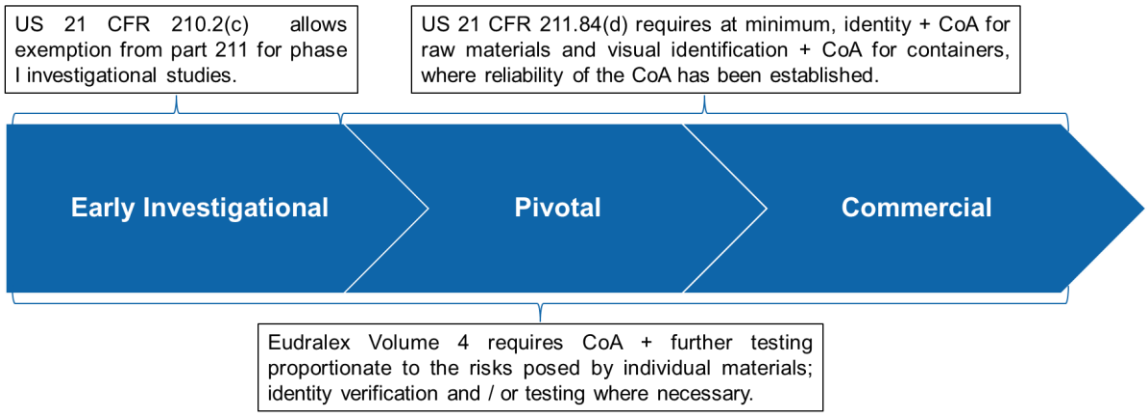

Figure 3.2-1 Application of United States and the European Union GMP Regulations related to material Identification Testing

Material Sampling and Retain Requirements

As part of the strategy for setting material specifications, the requirements for material sampling and retention in routine operations should be considered. According to EudraLex, Vol. 4, specification documents are required for all incoming materials and should include complete instructions for sampling and testing, as appropriate, as well as retention of samples for analytical purposes. The sampling and retain strategy should consider the risks and limitations related to the specific product. The use of dedicated raw material vials or containers for quality control testing is generally recommended to avoid contamination or impact on material by additional freeze-thaw cycles, in the case of raw materials that are stored and shipped frozen. For raw materials, retention samples of critical raw materials are important for investigation of product quality issues, whereas retention of starting materials or intermediates may be challenging due to limited quantities available beyond that needed for manufacturing. These intrinsic limitations are justifiable where they may impact the availability of the finished product; in such cases, appropriate mitigation measures should be applied and the strategy documented.

Considerations for Investigational/Early Clinical Stage (Pre pivotal)

Besides the general considerations mentioned here, there are considerations for investigational/early clinical stage programs. The list includes but is not limited to:

- Material must allow manufacturing of a functional product.
- Under EU regulations, reliance on the documentation provided by the supplier (e.g., CoA, CoO, TSE/BSE certificate) is acceptable, and no physical or chemical identification testing must be performed if the ATMP manufacturer fully understands the associated risks and has appropriate risk mitigations in place (e.g., supplier qualification, Quality Assurance Agreement) ***(1, 6)***.
- Under U.S. regulation, reliance on supplier documentation is acceptable only for Phase I investigational studies ***(11)***.
    - By the time sponsors start a Phase 2 study, FDA expects that manufacturers will operate under the GMPs outlined in 21 CFR 211, Subpart E).
    - A sponsor may rely on vendor information for confirmation of purity, strength, and quality if they perform a specific identity test and have confidence in the supplier (e.g., prior history of quality assurance and testing verification, risk assessment as part of vendor qualification). The risk-based approach to material qualification for ATMP is also

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outlined in USP <1043>. For example, Tier 1 Ancillary Material relies on the vendor's CoA without confirming the test results for FDA-approved products utilized as raw materials.

- Components may be received on verification of vendor documentation, such as a Product Certificate or Certificate of Compliance. Primary container–closure systems should, at a minimum, be subject to quality control visual examination.
- Perform incoming testing of identity for critical materials, for example, full sequencing on receipt of any plasmids and viral vectors used for ex-vivo gene modification.
- Avoid using RUO material. If use of an RUO material is inevitable, perform a risk assessment in terms of functionality and safety, for example:
  - Functionality (functional attributes) of the lot can be tested through an engineering run prior to the clinical manufacturing of cell and gene therapies.
  - Based on the risk assessment for the raw material, mitigate the risks as needed, for example, additional testing for identity, functionality, and safety, or add sterility control prior to use for removal of microorganism (e.g., sterile filtration, gamma radiation, heat treatment).
  - Use the same lot, if applicable, for manufacturing toxicology-study sample and Phase 1 clinical trial material to minimize the risks for lot variation or unknown risks of the RUO material.
- Selection criteria for excipients should be established and documented in the early phase of product development.
  - For cell-based products, excipients may be selected that support stability and functionality of the product cells; these may include culture media, electrolyte solutions for injections, exogeneous proteins, human serum, or cryoprotectants (USP <1046>) ***(12)***.
  - For vector-based products, final formulation depends on the physiochemical and biological characteristics of the vector system; ionic buffers, organic carbohydrates, amino acids, surfactants and extraneous proteins are common excipients (USP <1047>) ***(13)***.
- For non-approved and novel excipients, a complete safety assessment should be performed as described in USP <1046> and should include risk-benefit assessment according to FDA *Guidance for Industry: Nonclinical Studies for the Safety Evaluation of Pharmaceutical Excipients*.

Considerations for Late Stage/Pivotal Clinical Trials

Implementing the following aspects is recommended for commercial manufacturing:

- Limited or no amount of lower-grade materials (e.g., RUO).
- Material attribute assessments based on product and process development data (This will help to understand lot-to-lot variability and to define second supplier.)
- Limited or no amount of human- and animal-derived materials, if possible; if these materials cannot be avoided, the highest possible grade should be used.
- An internal material specification has been defined, that is based on a thorough material attribute assessment and reflects the specific needs of the product. Material specifications at this stage should not be a copy of supplier specifications unless further justified.
  Note: Vendor or supplier specifications often do not reflect the intended use in ATMP manufacturing (e.g., cytokines).

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- Analytical capabilities to allow for internal material release-testing should be established; each parameter according to the internal material specification must be tested internally, including contract labs.
  Note: This effort might be very resource-intensive and should be planned carefully). For material qualification, tests are performed prior to the process validation. A minimum of three lots will be required to complete this. If material qualification cannot be performed prior to process validation, or if sourcing three lots is not possible prior to the usage of materials for commercial GMP purposes, concurrent qualification can be used if it meets the following requirements:

  1) Quality approval on the use of concurrent qualification.
  2) At least one lot must pass all testing.
  3) Qualification of previously qualified material is leveraged if (a) they are from the same manufacturer and have different part numbers, and the change is due to processing; (b) a technical assessment shows equivalence between the new and previously qualified material; and (c) the supplier is on the approved supplier list.

- When using medical devices and approved medicinal drugs as materials, in case of limited inventory (drug shortage), it should be considered that material allocation could go to other entities (e.g., hospitals, clinics).
- Prior to the start of late-phase (pivotal) manufacturing, process-consumable-related risk assessments should be used to identify and evaluate the risks related to each specific scenario of usage, for each component, with consideration for such process parameters as temperature and duration of contact, evaluated against material and component attributes such as materials of construction and composition of the process stream.
- Components that have contact with the process stream should undergo quality control visual examination and/or testing, based on the specific risks identified through assessment. Primary container closure systems should be tested, at a minimum, for sterility, integrity, and bacterial endotoxin, in addition to visual examination, prior to usage.
- Leachables and extractables and/or chemical compatibility might be challenged (e.g., for DMSO-containing formulations), and study data might be necessary to demonstrate safety. Testing should aim to establish the worst-case drug-product profile to facilitate development of specifications and acceptance criteria where needed.

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Q 3.3. What are specific considerations for ATMP single-use components and final containers?

For control of single-use components in all phases of clinical manufacturing, the manufacturing site and supplier for each component should be identified. Full traceability of the supply chain should be known to identify any potential entry routes for adulterants or contaminants. Well-characterized, sterile, pyrogen-free components with intended usage as injectable/implantable medical devices, as well as drug delivery devices, present the lowest level of risk for usage within manufacturing processes.

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Topic 4. Risk-Based Supplier Qualification

ATMP material-supplier qualification is a risk-based approach. The level of oversight should be proportionate to the risks posed by the individual materials, taking into account their source, manufacturing process, supply chain complexity, and final use to which the material will be put in the ATMP.

Note: For products that fall under PIC/S GMP regulation, Annex 2A provides additional requirements.

Besides general requirements, which are applicable to all clinical stages and commercial manufacturing, several requirements differentiate between early- and late-stage clinical manufacturing to commercial manufacturing.

Q 4.1. What are the key considerations when qualifying an ATMP material vendor or supplier?

General Considerations

- Material manufactured under GMP regulation
  - Paper audit (e.g., questionnaire) may be acceptable for lower-risk raw material, such as off-the-shelf material used in the upstream process of cell and gene therapies (early stage only).
  - Whether an on-site audit needs to be performed at a manufacturer's or distributor's premises should be defined based on QRM principles. Generally, audits need to be performed at vendors of all materials defined as critical for the manufacturing process according to its product risk profile.
  - Per PIC/S, Annex 2A, a regular qualification of the vendors (e.g., manufacturers, distributors) of all materials to confirm that they comply with the relevant GMP requirements should be performed. This regulation may not apply to one-time-use materials such as a master cell bank.
- Material manufactured under non-GMP conditions
  - SME should conduct a technical visit to evaluate the risks, in case an audit is not feasible (or accepted) for the supplier, such as a standard supplier with a low-quality management system, for example, one using off-the shelf RUO material.
- If, during technical visits or audits, gaps are identified, the sponsor should evaluate the risks of using the material and provide a justification in terms of safety and functionality by means of additional testing (e.g., product in-process testing, performance test, use of the same lot for manufacturing of toxicology-study sample and Phase 1 clinical-trial material).

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- For U.S. suppliers of approved or licensed therapeutic products, reports of past audits may be leveraged along with cross-reference of the Device Master File in lieu of a technical audit. Standard risk-reduction measures should still be applied to the receipt and release of these materials ***(14)***.
- A written agreement for change notifications from the supplier of the critical raw material is recommended to be in place regardless of the clinical phase to evaluate potential impact." (See **Q1.1** and **Topic 3** for the definition of critical raw material).

Following the guidance of PIC/S, Annex 2A, Section 5.27, is recommended, as it states:

"For starting material of human origin, the agreement between the ATMP manufacturer (or, as appropriate, the MAH) and the supplier (including blood and tissue establishments) should contain clear provisions about the transfer of information. In particular, this should include test results performed by the supplier, traceability data, and transmission of health donor information that may become available after the supply that may have an impact on the quality or safety of the ATMPs manufactured…." ***(6)***.

Specific Consideration for Individualized Therapies
For individualized therapies (e.g., autologous cell therapies), since each batch of product is manufactured for a specific patient, an error-free system for patient-derived materials (e.g., tumor samples or leukocytes) must be in place to maintain the chain of identity and chain of custody.

Early Clinical Stage

- A quality system should be in place. The level of quality standard should be adequate to the criticality of the raw material. To enable a clinical trial application, the following minimum requirements must be met:
  - Supplier quality system must be capable of providing safety-relevant documentation.
  - Traceability of materials and reagents (e.g., a unique material or catalogue number assigned).
  - Description of the origin and history of the material (e.g., donor tissue, chemically defined or recombinant, based on a cell bank).
  - For EU-regulated starting materials, GMP principles should be applied ***(5)***.
- For customized materials (e.g., viral vector, custom-made media and buffers), a quality agreement should be in place.
- For off-the-shelf or low-risk materials, it may be sufficient to have a change-notification agreement in place before comprehensive quality agreements are in place.

Late Clinical Stage and Commercial

- The quality system standard should be increased to meet GMP regulations or GMP principles defined by EMA whenever possible. The level of GMP adherence should be justified based on the risks of the gaps, with the understanding of material attributes and impact on manufacturing the final drug product.

Note: Material grades are not standardized. The term "GMP grade" does not necessarily mean that material is manufactured and distributed under current GMP regulations. Instead, the supplier's GMP should be confirmed through assessment by questionnaire or audit to verify the material is of GMP grade.

- To ensure supply for confirmatory or pivotal clinical trials and the approved product, establishing second suppliers of materials, especially critical raw materials, is essential. Where high-risk materials (e.g., animal-derived, high-risk supplier) are determined to be critical to product availability and cannot be eliminated or substituted, risk assessments should be utilized to identify

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and evaluate the specific risks posed by the material in relation to the production and availability of the finished product. Consideration of such factors as the number of suppliers and/or alternatives available, especially where materials are single-sourced, as well as the lead time and shelf life, for example, are crucial to ensuring the control measures applied are commensurate with the level of risk posed. Specific control measures sufficient to eliminate or mitigate to an acceptable level each risk scenario identified should be established, documented, and applied to each receipt of the respective material.

- Having a second qualified supplier is preferred to reduce the risk of drug shortages and ensure business continuity.

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Topic 5. Lifecycle Management Considerations for Commercial Manufacturing

Ideally, lifecycle management of raw materials starts early in the development of an ATMP product. It evolves as the product moves through the early clinical phases, through commercialization, and continues throughout the lifecycle of the product as more data and understanding are gained.

When an ATMP product is commercialized, the raw materials should be reviewed from the perspective of lifecycle management and potential new industry guidance. New information may be available that will change the risk-rating for raw materials and, therefore, testing and/or mitigation requirements. Topics to consider include:

- Raw material that was single-sourced may now have additional suppliers that can be considered as a secondary or tertiary source.
- Suppliers may now offer a raw material as GMP grade that was previously labeled for RUO.
- Historical data will now be available on raw materials that may allow for updates to the incoming testing requirements and reassessment of testing specifications, including the widening of specifications.
- Historical data will now be available for reassessment of a supplier's qualification that may allow for reduced testing on the receipt of raw materials.

Product regulatory filings are important to consider with any raw material reassessment, as it may require updates in multiple markets. The timing of these regulatory submissions should also be considered from the perspective of how it will impact the manufacturing and planning schedule and, by extension, patient supply.

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Q 5.1. How do you manage the situation when the supplier does not provide GMP service for the material, for example, if only Research Use Only grade is available?

The sponsor (i.e., cell-therapy drug manufacturer) can identify the risks on-site and figure out the pathway for enabling the material to be fit for purpose. The following case study will provide an example of quality risk management.

Case Study: Example of Implementation of GMP Principles for ATMP Raw Materials

This fictional case study is about raw materials used in genome-editing for induced pluripotent stem cells (iPSCs), which is an upstream cell-engineering process followed by single-cell cloning and, finally, differentiation to a CAR-T cell drug product. Health authorities give allowance for manufacturing materials, such as genome-editing materials, generation of iPSCs, and the subsequent selection of process cells, under the principles of GMP operation rather than full GMP compliance ***(5, 6, 15)***. In Japan, PMDA does not specify GMP or non-GMP adherence during the clinical phase, and the clinical-study sponsor owns the responsibility to secure the product quality for patients.

This case study is an example of the implementation of "Principles of GMP" for gRNA and nuclease used in the genome-editing of iPSCs. In this context, principles of GMP refer to the application of specific GMP elements, such as robust documentation practices and contamination-control measures, even when a fully GMP-compliant environment is not available.

Statement: Certified GMP facility and its established GMP-compliant quality management system is not available for a kind of custom-made small-volume production at the supplier. The only existing development laboratories can be used for manufacturing a product of appropriate quality for fit-for-purpose.

Approach: Instead of performing a normal GMP audit, strict sponsor's Quality team oversight is required on gap identification and definition of control measures. After those measures are completed, the sponsor's Quality confirms readiness on site prior to initiation of the contract manufacturing work.

Agreed measures are spelled out in a Scope of Work (or ideally, a Quality Agreement) with the signatures of both parties.

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Q 5.2. How to implement principles of GMP and enable IND filing?

- Identify gaps on site and address them by measures for upgrade to principles of GMP. See contamination control strategy below as an example.
- Provide detailed and complete production records with sufficient information for enabling an IND/BLA filing, which is traceable for all aspects of the manufacturing process and any incidents.

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Q 5.3. How to mitigate risks for cross-contamination and microbial contamination?

- As a minimum requirement, use a non-ISO-grade certified facility (i.e., controlled research and development (R&D) laboratory) but dedicate its use for one product at a time. Replace all tubing of instruments with brand new tubing dedicated for the contract work.
- Make the bulk product in the controlled R&D lab and transfer it into a qualified isolator for sterile filtration and fill–finishing. The adequacy of the aseptic operation and environmental control is confirmed through aseptic process simulation before actual production.

Note: The supplier will not commit to producing "sterile" product; however, the sponsor is confident with the patient safety-based assurance on the material control, downstream manufacturing control, and overall quality control strategy of the final drug product.

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Q 5.4. How to assure the quality of the product (i.e., gRNA, nuclease)?

- Define the material specifications and perform quality control release-testing at a GMP-compliant laboratory.
- Store the product in a GMP-controlled warehouse once it is manufactured.
- Conduct a stability study to ensure the quality of the raw material during storage when no stability data is available from the supplier.

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Points to Consider No. 13: Materials in ATMP Manufacturing

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